Contents

General safety information

Armstrong EvaPack™ humidifier control cabinet must be installed by qualified personnel using this manual as a reference.

Always seek the advice of a specialist when selecting or installing equipment. Contact Armstrong International or one of its local representatives for further Information.

IMPORTANT

Please read and observe the enclosed safety information, as well as the warning labels affixed to the inside of the humidifier, before installation or maintenance.

WARNINGS AND SAFETY SYMBOLS

SAFETY WARNINGS AND SYMBOLS USED IN THIS MANUAL	
	**Danger! Caution.** General safety instructions, violation of which could lead to malfunctions and/or personal injury and/or property damage.
	**Danger! High voltage.** If high voltages are present inside the device or any of its components, failure to heed this warning may result in death or serious injury to the user. people and/or major equipment malfunctions.
	**Danger! High temperatures.** Make sure that protective equipment is worn and keep a safe distance from the machine. between the device and any materials likely to be damaged by heat.
	**Electrostatic hazard.** Device components may be subject to damage as they are highly sensitive to electrostatic discharge.
	**Möbius strip.** Some parts of the device can be recycled. The user is responsible for their disposal. Follow the recycling recommendations adapted to the materials used in your area.

TRANSPORT AND STORAGE

If your packages are damaged or missing, any complaint must be made to the carrier by registered letter with acknowledgement of receipt within 24 hours, and reported to Devatec or its representative.

The humidifier should be stored in a dry, frost-free place, protected from shocks and vibrations.

GENERAL

This manual contains all the details you need for planning and installing the EvaPack™ control cabinet. Start up, operation and maintenance are clearly indicated.

This manual is intended for engineers and technical personnel. Maintenance, service and repairs must be carried out by qualified and competent personnel and are the responsibility of the customer.

Risks or hazards, especially when working at height, must be identified by an authorized and competent health and safety officer. A safety perimeter must be set up.

The distributor shall not be liable for any injury or accident caused by inattention, negligence or improper operation, whether deliberate or not.
Always disconnect the power supply before carrying out maintenance.

INTENDED USE

The control units are intended exclusively for humidifying air, in air handling units or in room environments, within the limits of the specified operating conditions. Any other use is considered improper and may render the system unsafe.

ELECTRICITY

All work on the electrical installation must be carried out by a qualified technician. The customer is responsible for this work. It is the installer's responsibility to provide the correct cable cross-section and circuit breaker protection. The control cabinet must be earthed using a suitable conductor.

NOTICE

Armstrong International reserves the right to change the specifications or design of equipment described in this brochure without notice.

Declaration of conformity

APPLIED GUIDELINES

Electromagnetic Compatibility (EMC) Directive:	**89/336/EEC, 2004/108/EC**
Low Voltage Directive:	**73/23/EEC, 2006/95/EC**
Machine Directive:	**98/37/EC, 2006/42/EC**

THE DEVICE COMPLIES WITH

EN 61000-6-3: EMC: Generic standard: Emission in residential environments.
EN 55022 class B; Conducted and radiated emission.

EN 61000-6-2: EMC: Generic standard: Immunity in industrial environments.
EN 61000-4-3: Radiated immunity, Electromagnetic fields.
EN 61000-4-6: Line immunity, Radio-frequency line disturbance.
EN 61000-4-4: Immunity to fast burst electrical transients.
EN 61000-4-5: Immunity to shock waves.
EN 61000-4-2: Immunity to electrostatic discharge.

EN 60335-1: Low voltage: Safety of household and similar electrical appliances.

EN 60335-2-88: Low voltage: Safety of household and similar electrical appliances, in particular humidifiers and control cabinets.

EN 60204-1: Safety of machinery - Electrical equipment of machines.

MANUFACTURER'S NAME AND ADDRESS

Devatec SAS
185 Boulevard des Frères Rousseau
76550 Offranville - FRANCE

EQUIPMENT TYPE
Humidifier control panel

MODEL NAME AND SERIES
EvaPack™ control cabinet

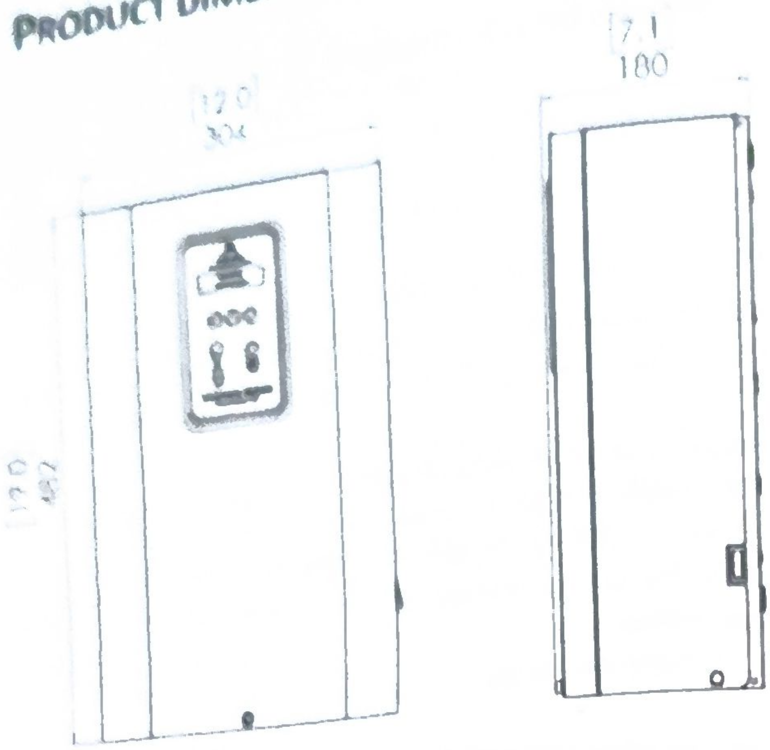
YEAR OF PRODUCTION
2016

We hereby declare that the equipment specified above conforms to the directives mentioned at the beginning of this declaration.

Name: FRAMBOT Jean-François
Position : General Manager
Date: 30/09/2024
Signature:

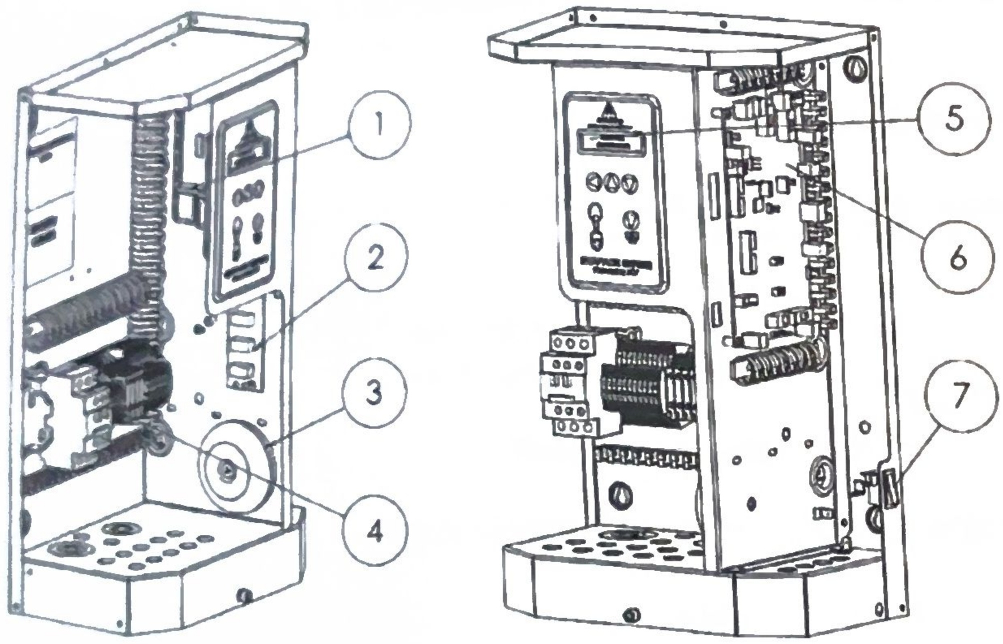
Product information

PRODUCT DIMENSIONS



	Control cabinet
	Surface area [mm²]
Minimum recommended cable cross-section	1.5

	Mass [kg]
Operating weight	12
Packed weight	13



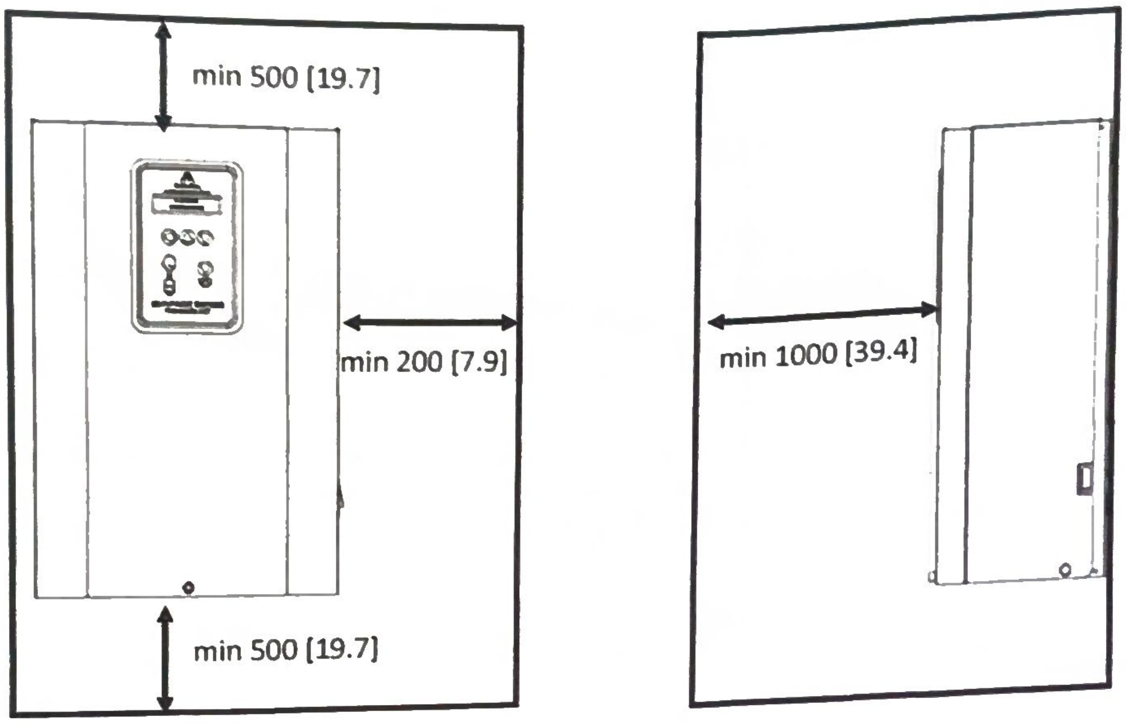
1	Level detection board
2	3-relay board
3	Toroidal transformer
4	Connection terminals
5	Display board
6	Main board
7	Switch ON/OFF

Product installation

The control cabinet must be installed in an environment with a temperature between 5°C and 40°C. Relative humidity must not exceed 80%.
The control cabinet is designed for wall mounting. Please ensure that the support material can bear the weight of the control cabinet.

POSITIONING THE UNIT

To facilitate installation and maintenance, please follow the dimensional drawings below:
Dimensions in mm [in]

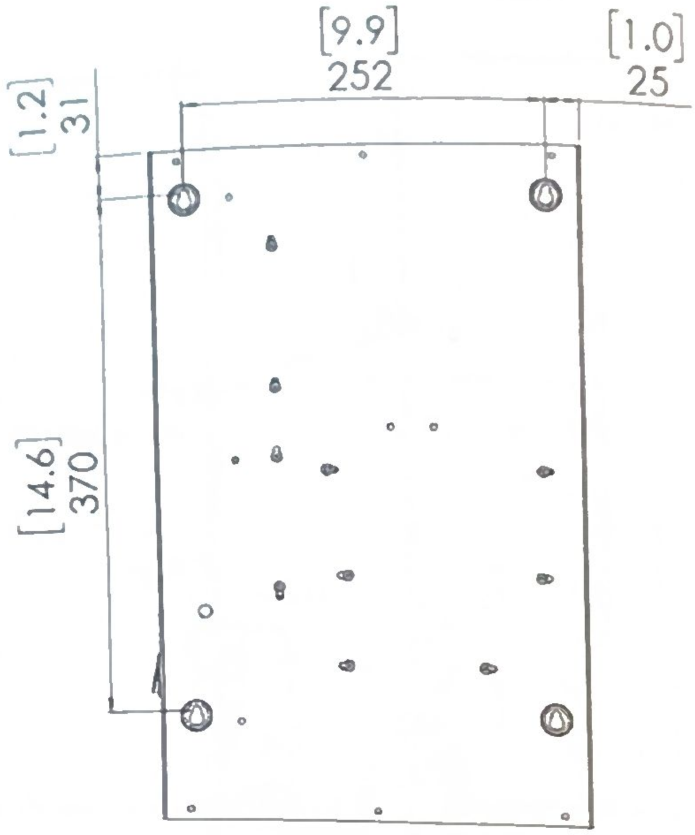


FIXING THE UNIT

A drilling template is supplied in the carton, so be sure to remove it before discarding the packaging.
Be sure to use a fastening system that is suitable for your surface.

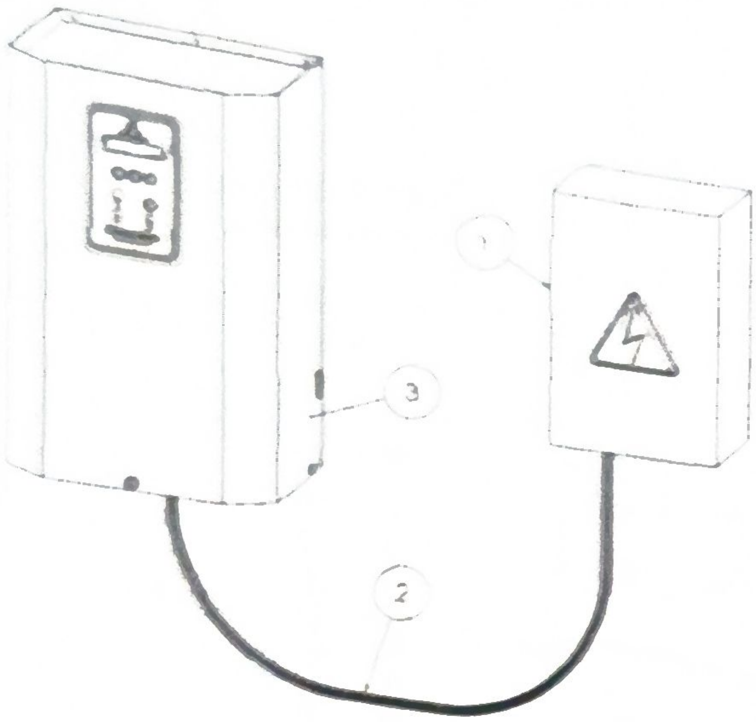
Take the drilling template supplied with your appliance, mark and drill in the indicated places (holes according to the dowel selected in relation to the support materials).

Insert the dowels into the holes. Screw the top screws into the dowels, leaving them protruding by about 10mm.
Hang the appliance on these screws (top) and screw in loosely.



Dimensions in mm [in]

Electrical connection



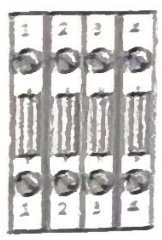
1 - Power supply and circuit breaker (close to unit)
2 - Power supply cable (Ø: 1.5 mm²)
3 - Electrical compartment

WARNING:
Install an earth leakage circuit breaker to prevent the risk of electric shock.

In the electrical cabinet, it is imperative to install an earth leakage circuit breaker (ELCB) at the head of the power supply network serving the humidifiers and their cabinets. If several humidifiers are installed, it is preferable to install one earth leakage circuit breaker per humidifier and per cabinet, to avoid cutting off the entire installation.

INTERNAL PROTECTION

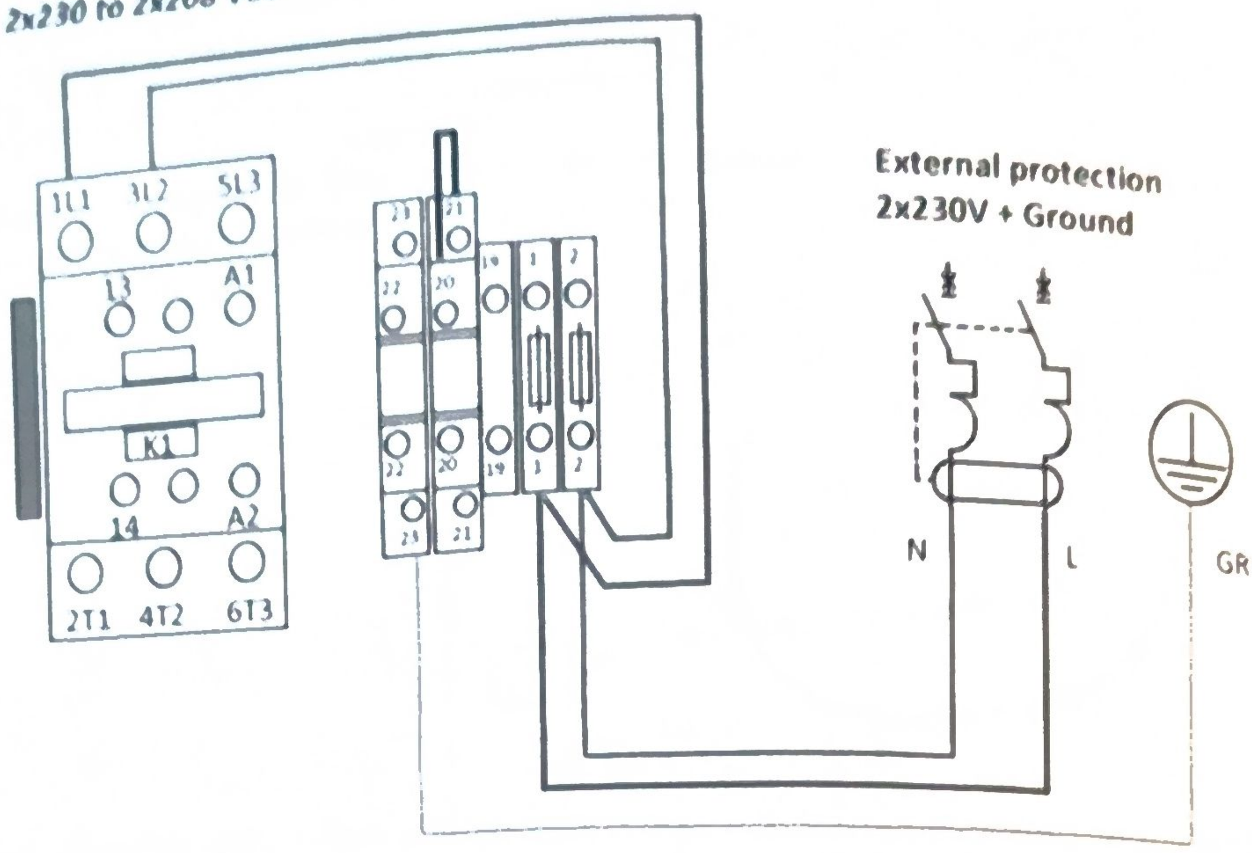
Evapack control cabinet



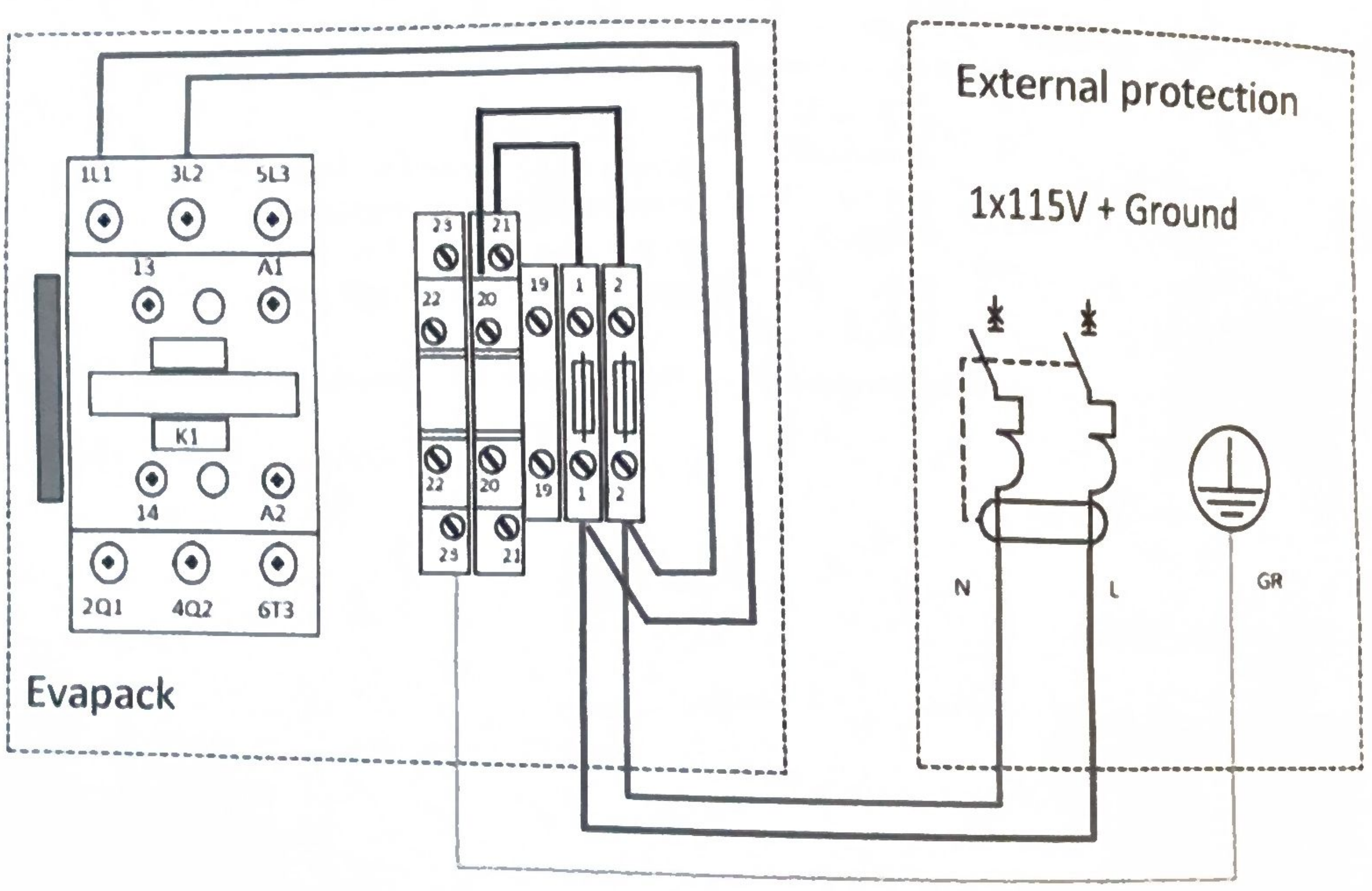
1: F1: 2 Amp / Timed / 5x20 mm
2: F2: 2 Amp / Timed / 5x20 mm
3: F3: 5 Amp / Timed / 5x20 mm
4: F4: 5 Amp / Timed / 5x20 mm

POWER SUPPLY

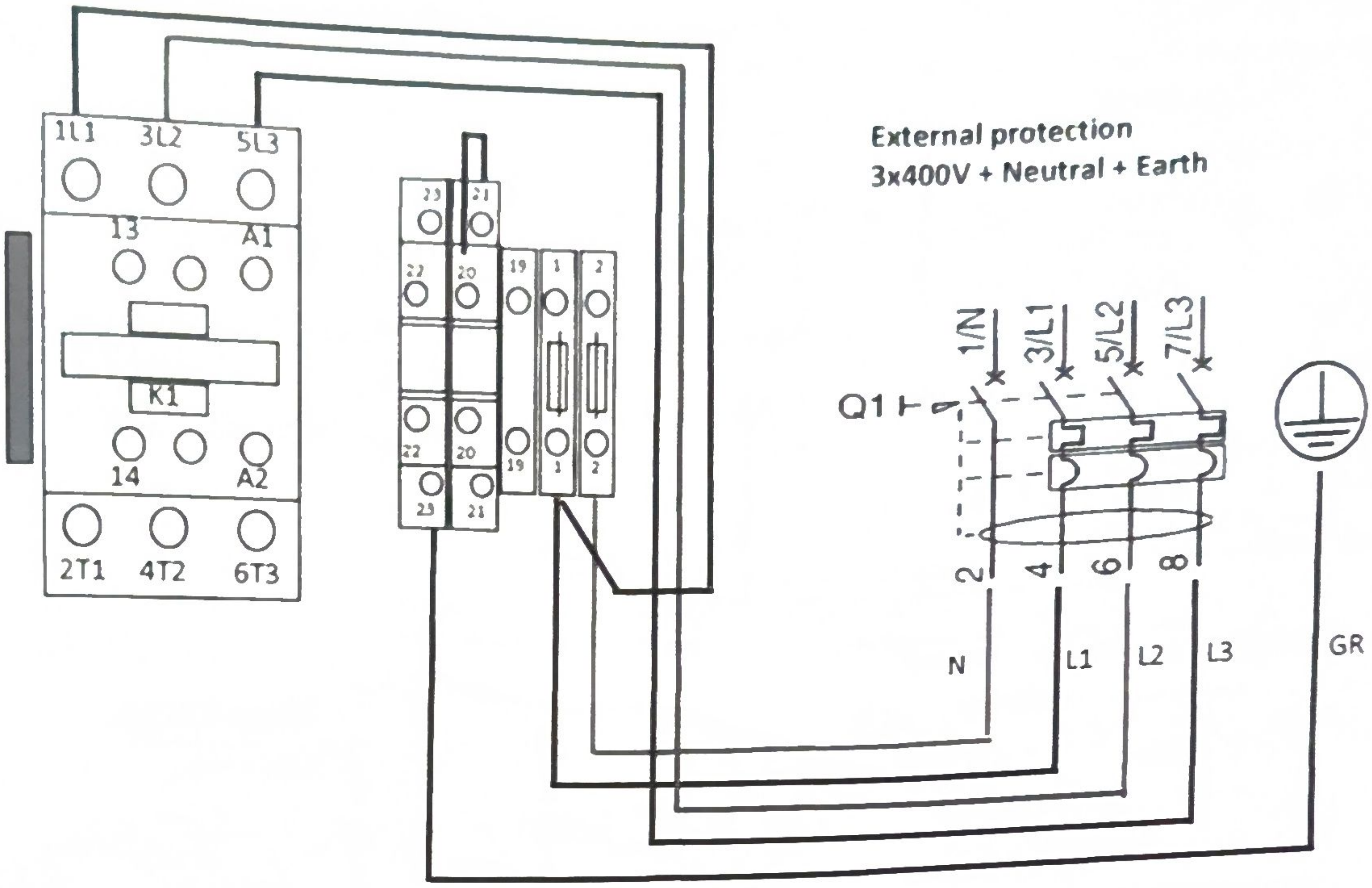
- *2x230 to 2x208 Vac*



- *1x115 to 1x110 Vac*



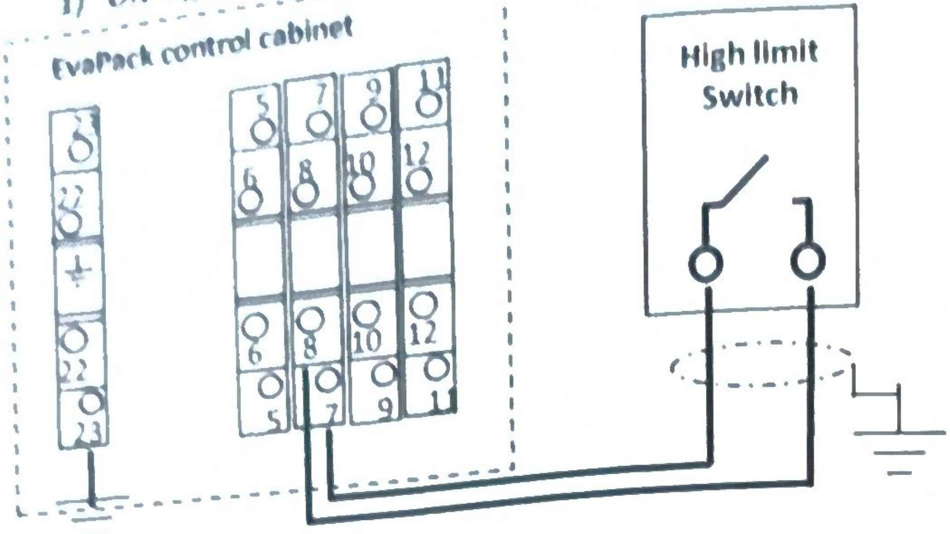
- *3x400V/3x380V + Neutral*



Control

Use shielded cable (recommended 0.75mm²) for control connection.

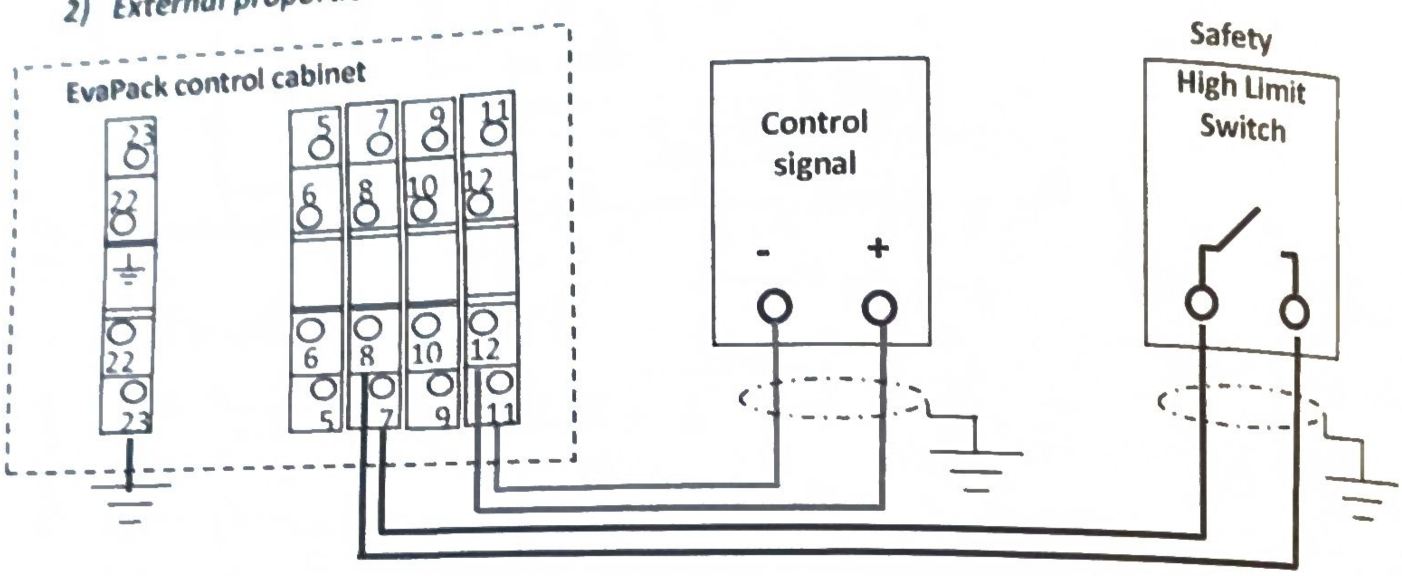
1) ON-Off control



Setting the control signal

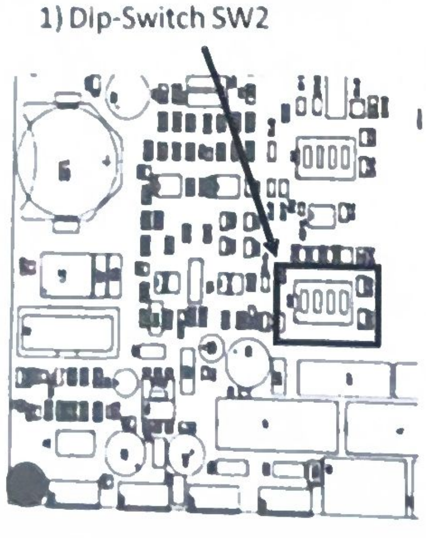
CTRL SIGNAL TYPE
ON/OFF CONTROL

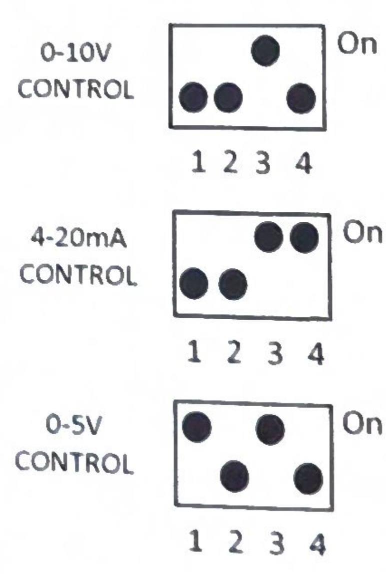
2) External proportional control



To configure

1) Dip-Switch SW2





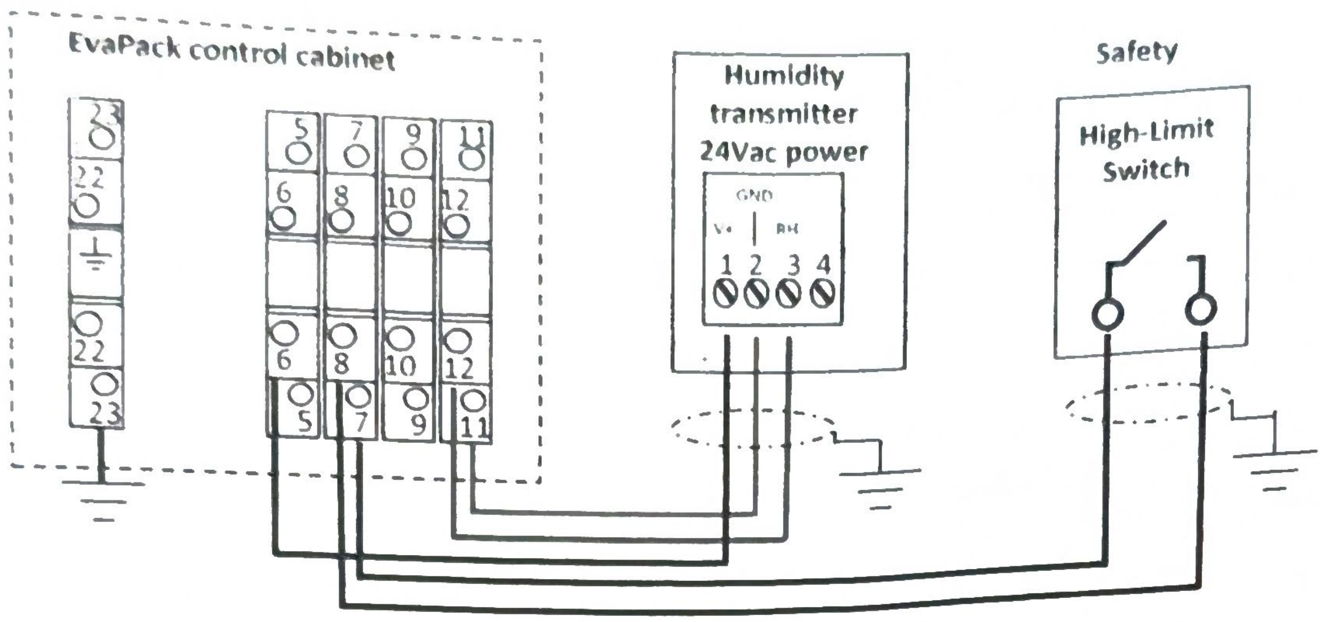
2) Set the control signal

CTRL SIGNAL TYPE
0-10V CONTROL

CTRL SIGNAL TYPE
4-20mA CONTROL

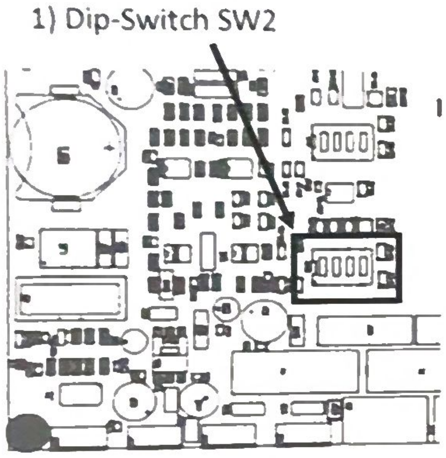
CTRL SIGNAL TYPE
0-5V CONTROL

3) Proportional control with humidity transmitter

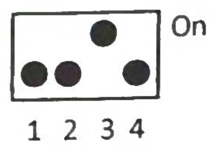


To configure

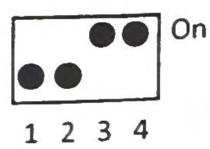
1) Dip-Switch SW2



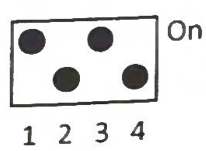
0-10V SENSOR



4-20mA SENSOR



0-5V SENSOR



2) Set the control signal

CTRL SIGNAL TYPE
0-10V SENSOR

CTRL SIGNAL TYPE
4-20mA SENSOR

CTRL SIGNAL TYPE
0-5V SENSOR

With digital control

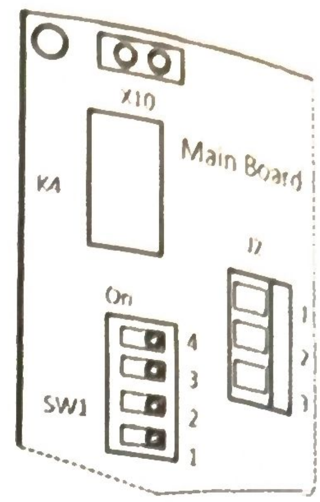
Control signal type setting

CTRL SIGNAL TYPE
DIGITAL CONTROL

CTRL SIGNAL TYPE
DIGITAL SENSOR

Connection: see next page

RS485-HARDWARE CONNECTION

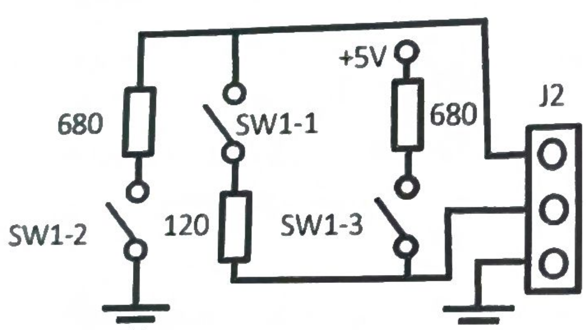


RS485 connection must be connected to connector J2 (main board)

- Terminal 1: Data -
- Terminal 2: Data +
- Terminal 3: Ground

DIP-switch SW1 is used to activate or deactivate the line resistor. In most cases, these resistors are useless and should be deactivated. (see connection diagram below right)

Fast communication	2400 to 9600 Bd
Package size	8 bits
Parity	No parity
Stop bit	1 or 2
Response time	2.5 seconds
Time between requests	Min: 100ms
Max. number of registers / request	5



1: Data -

2: Data +

3: GND

Modbus RTU communication parameters

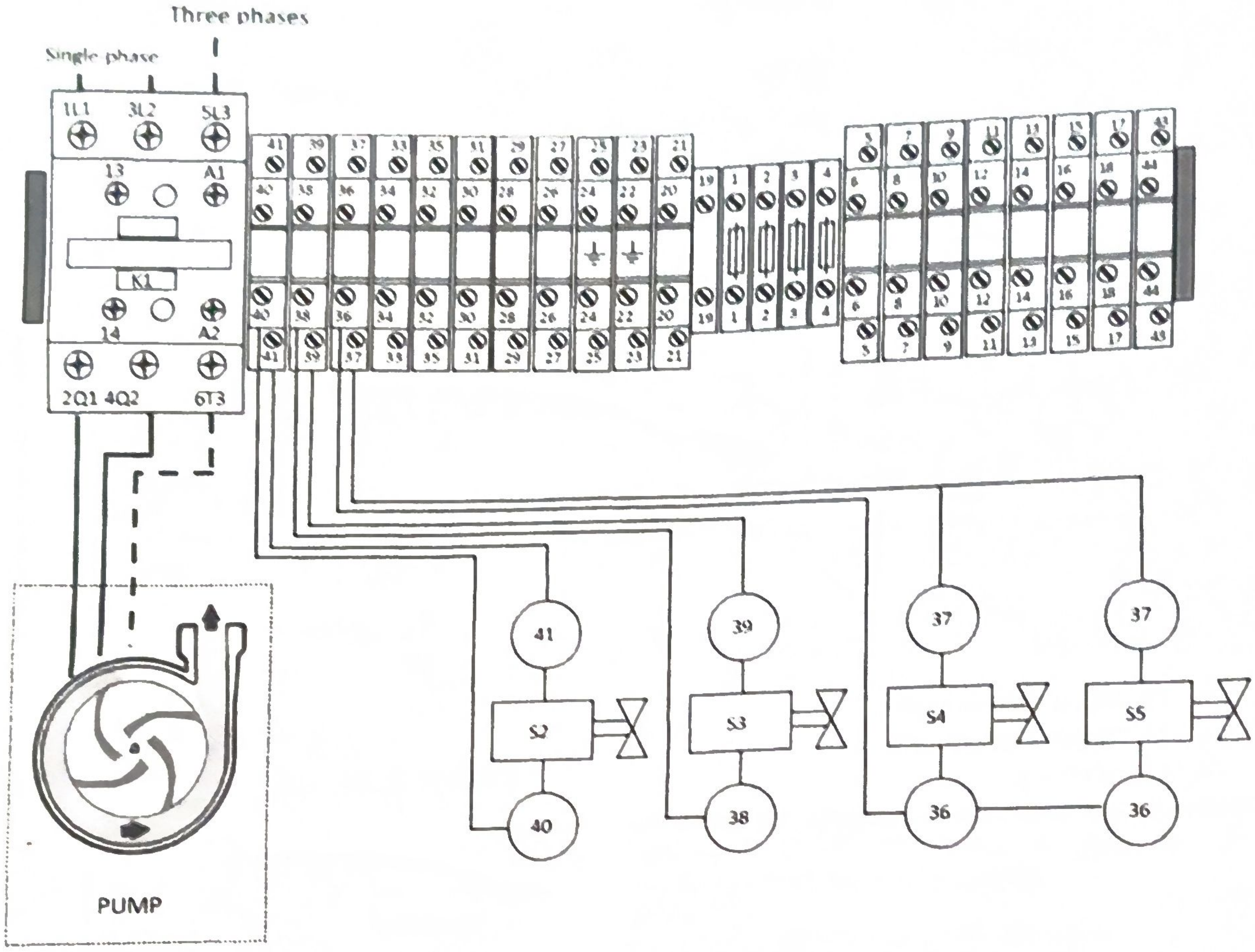
Description	Value	Modbus Address
Basin filling valve	0 => Filling valve: OFF I => Filling valve: ON	10001
Basin drain valve	0 => Basin drain valve: OFF I => Basin drain valve: ON	10002
Pump	0 => Pump: OFF I => Pump: ON	10003
Control valve 1	0 => control valve 1 : OFF I => control valve 1: ON	10004
Control valve 2	0 => control valve 2 : OFF I => control valve 2: ON	10005
Control valve 3 - 4	0 => control valve 3 - 4 : OFF I => control valve 3 - 4 : ON	10006
Cleaning the water supply circuit (optional)	0 => Cleaning valve: OFF I => Cleaning valve: ON	10007
Limit switch	0 => Switch open I => Switch closed	10008
High water level sensor	0 => High water level not reached I => High water level reached	10009
Low water level sensor	0 => Low-water level not reached I => Low-water level reached	10010
Conductivity control (optional)	0 => Correct water conductivity I => Water conductivity too high: drain	10011
E0 : Default pump	0 => Pump operating correctly 1 => Pump faulty	10012
E1: First filling fault	0 => the first basin filling is correct I => the initial filling of the basin takes too long	10013
E2: Fault filling	0 => the basin is correctly filled I => the basin takes too long to fill	10014
E3: Faulty reload	0 => Recharge time is correct I => Recharge time too long	10015
E4: Defining high performance	0 => Time is correct I => Time is too long	10016
E5 : Detection of defective levels	0 => Correct level detection I => Problem with level sensors	10017

Description	Value	Modbus Address
On/Off Unit via BMS	0 => Unit stopped / I => Unit running	1
Type Unit	10 => EVAPACK	30001
RegVersion	RegVersion XX	30002
State of progress	1 = Off 2 = Humidification 3 = Manual drain 4 = Service 5 = End of season 6 = Fault 7 = Cleaning sequence	30003
Demand	from 0 to 100	30004
Control signal value	10 x [V] or 10 x [mA] or [%] or [%].	30005
Total operating time (High value)	value in hours	30006
Total operating time (Low value)	value in hours	30007
Maintenance to be carried out in :	value in hours	30008
Counter between 2 service intervals	value in hours	30009
End-of-season counter (EOS)	value in hours	30010
Type of water supply	1 = Tap water 2 = Softened water 3 = Partail DI water 4 = DI water	30011
Temperature 1	X : No temperature sensor or Display : XX [°C]	30012
Temperature 2	X : No temperature sensor or Display : XX [°C]	30013
Not used		40001
Not used		40002
Not used		40003
Digital demand	from 0 to 100%.	40004
Rh Set Point	10 to 99%Rh (default: 50%Hr)	40005
Adjustment service counter	1 to 200 x (100) hour	40006
Water supply type setting	1 = Tap water 2 = Softened water 3 = Partial water D 4 = DI water	40007
EOS drain control	0 = No 1 = Yes	40008
EOS drain timer	1 to 72 hours - Default: 72 hours	40009
Duration of cyclic draining	0 to 240 seconds	40010
Cyclic drainage frequency	1 to 60 min	40011
Water renewal cycle	1 to 48 hours	40012
Control signal setting	-10 à +10 %	40013
Temperature setting 1	-10 à +10 %	40014
Temperature setting 2	-10 à +10 %	40015
Proportional factor (PID)	0 à 50	40016
Integral factor (PID)	0 à 50	40017
Derivative factor (PID)	0 à 50	40018

Electrical installation

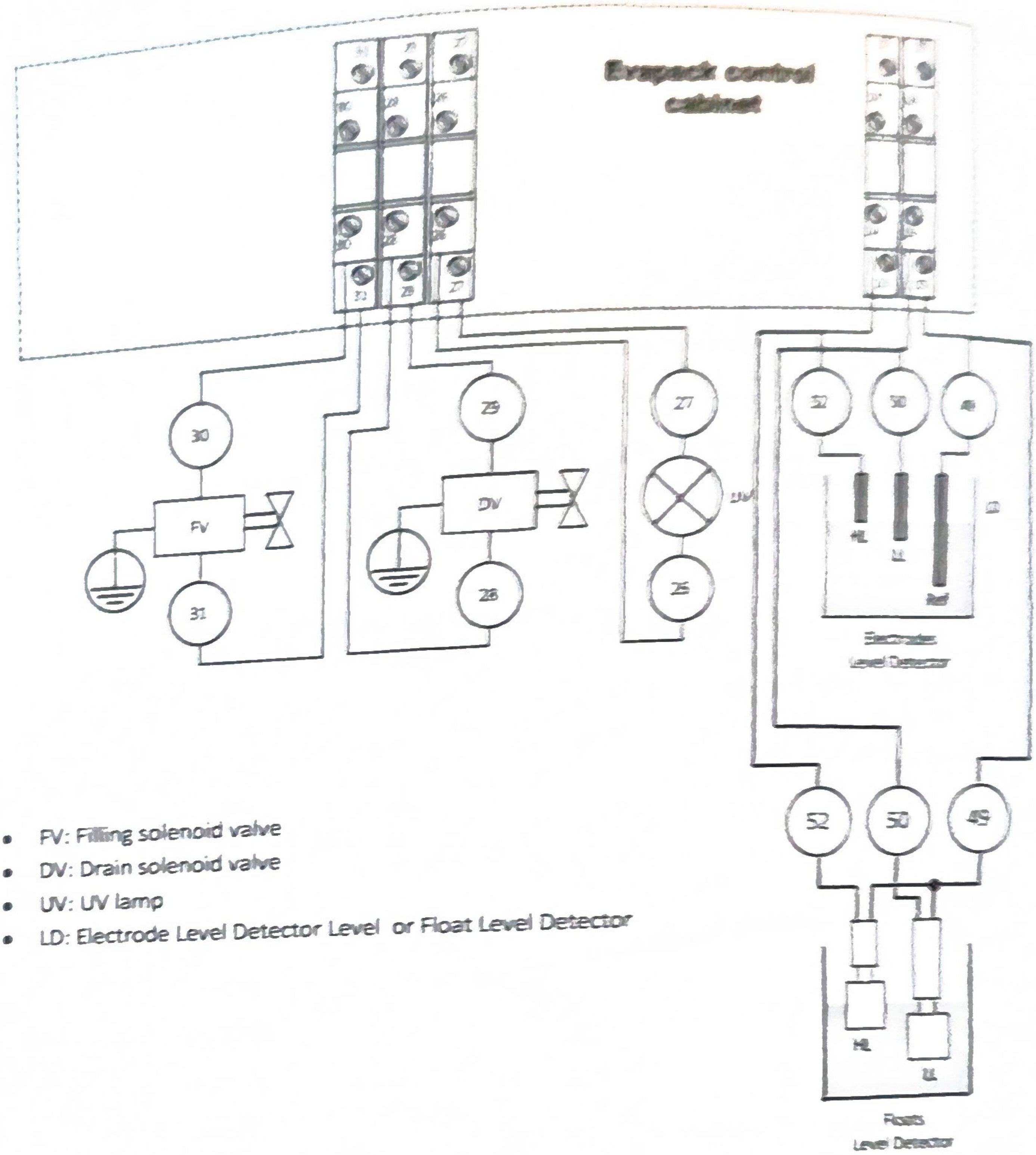
STEP CONTROL (PUMP + VALVE)

PUMP SUPPLY



- S2: Solenoid valve step control 2
- S3: Solenoid valve step control 3
- S4: Solenoid valve step control 4
- S5 : Solenoid valve step control 5

DRAIN VALVE, FILLING VALVE, UV LAMP, LEVEL SENSORS

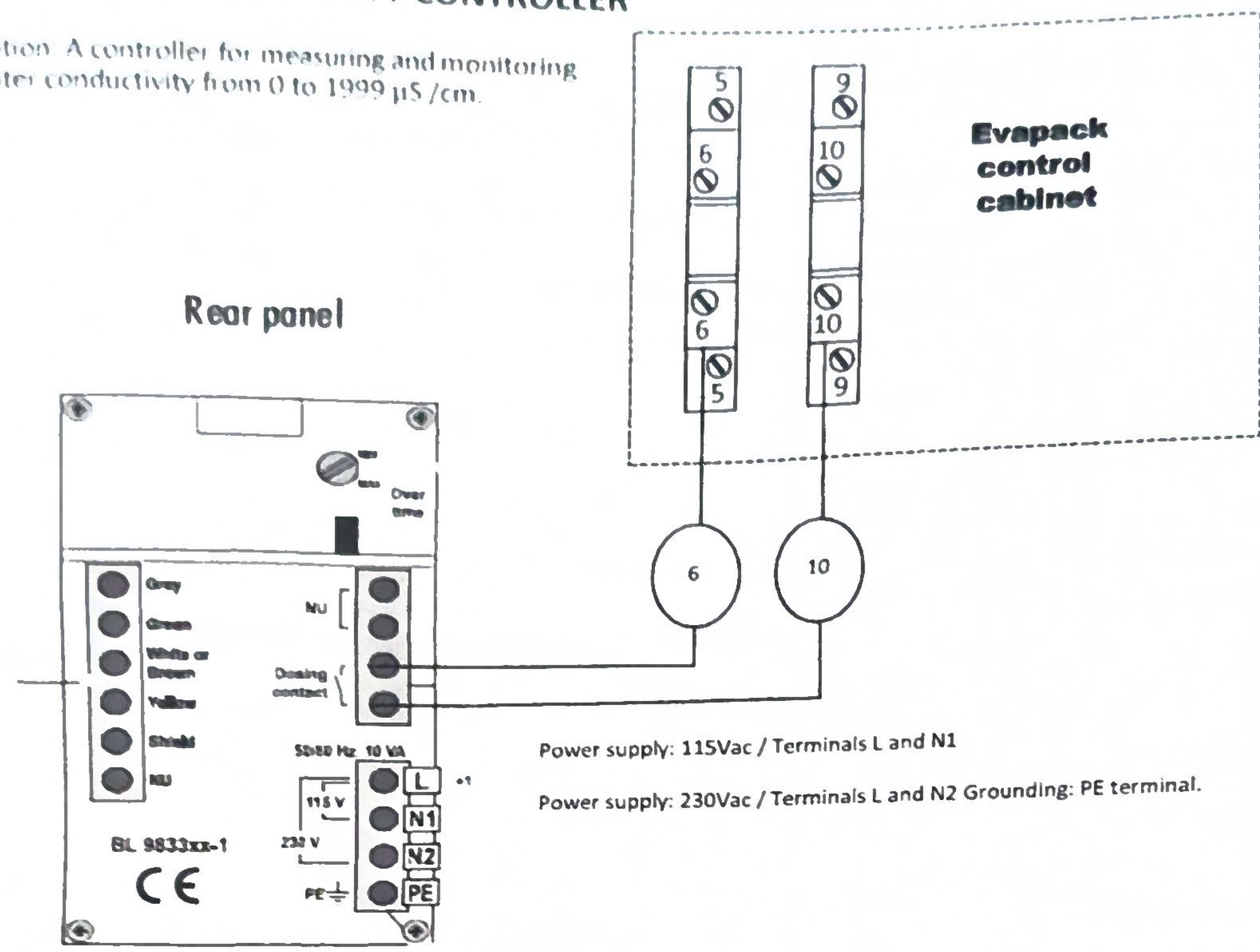


- FV: Filling solenoid valve
- DV: Drain solenoid valve
- UV: UV lamp
- LD: Electrode Level Detector Level or Float Level Detector

Options: Conductivity controller

Option: A controller for measuring and monitoring water conductivity from 0 to 1999 µS /cm.

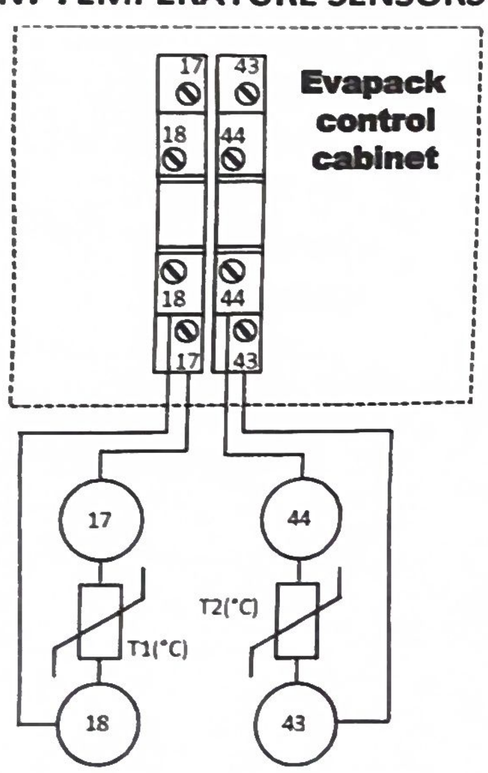
Rear panel



Power supply: 115Vac / Terminals L and N1

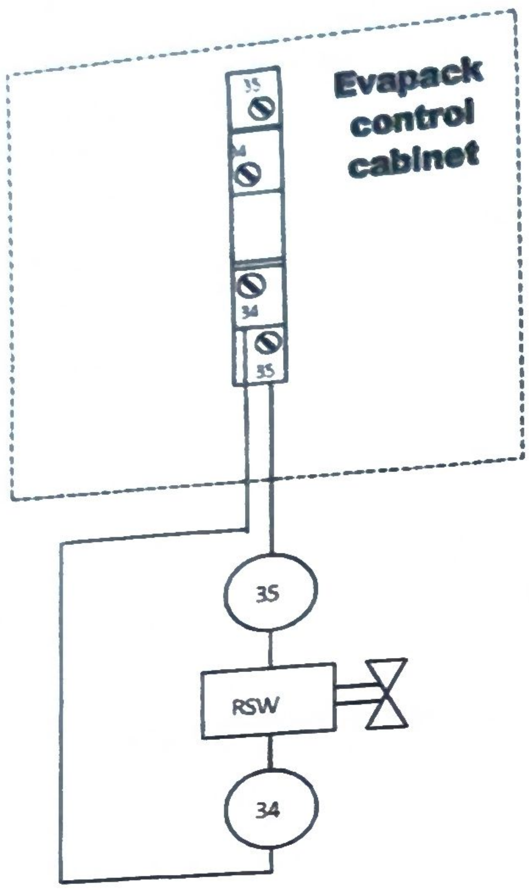
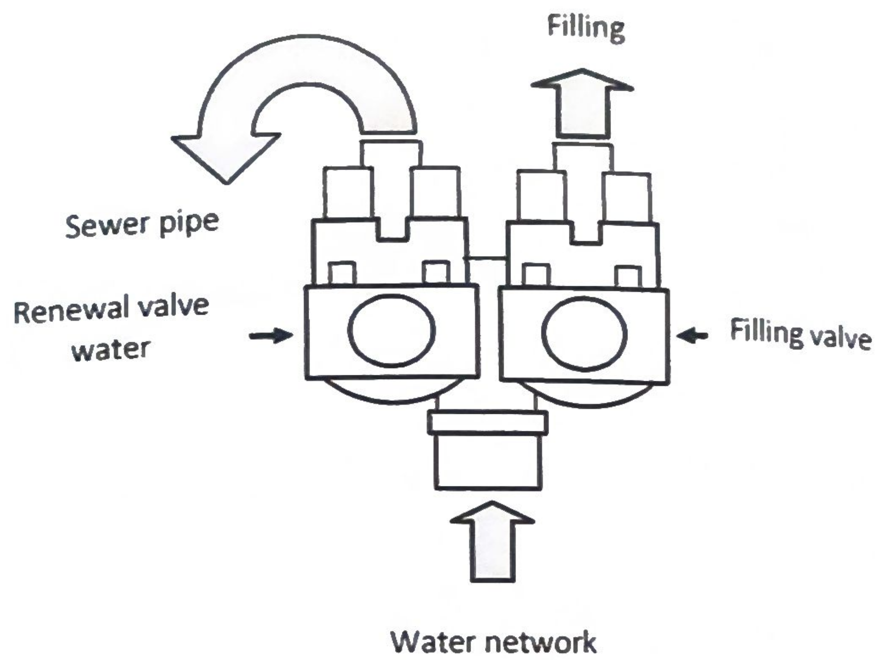
Power supply: 230Vac / Terminals L and N2 Grounding: PE terminal.

Option: Temperature sensors

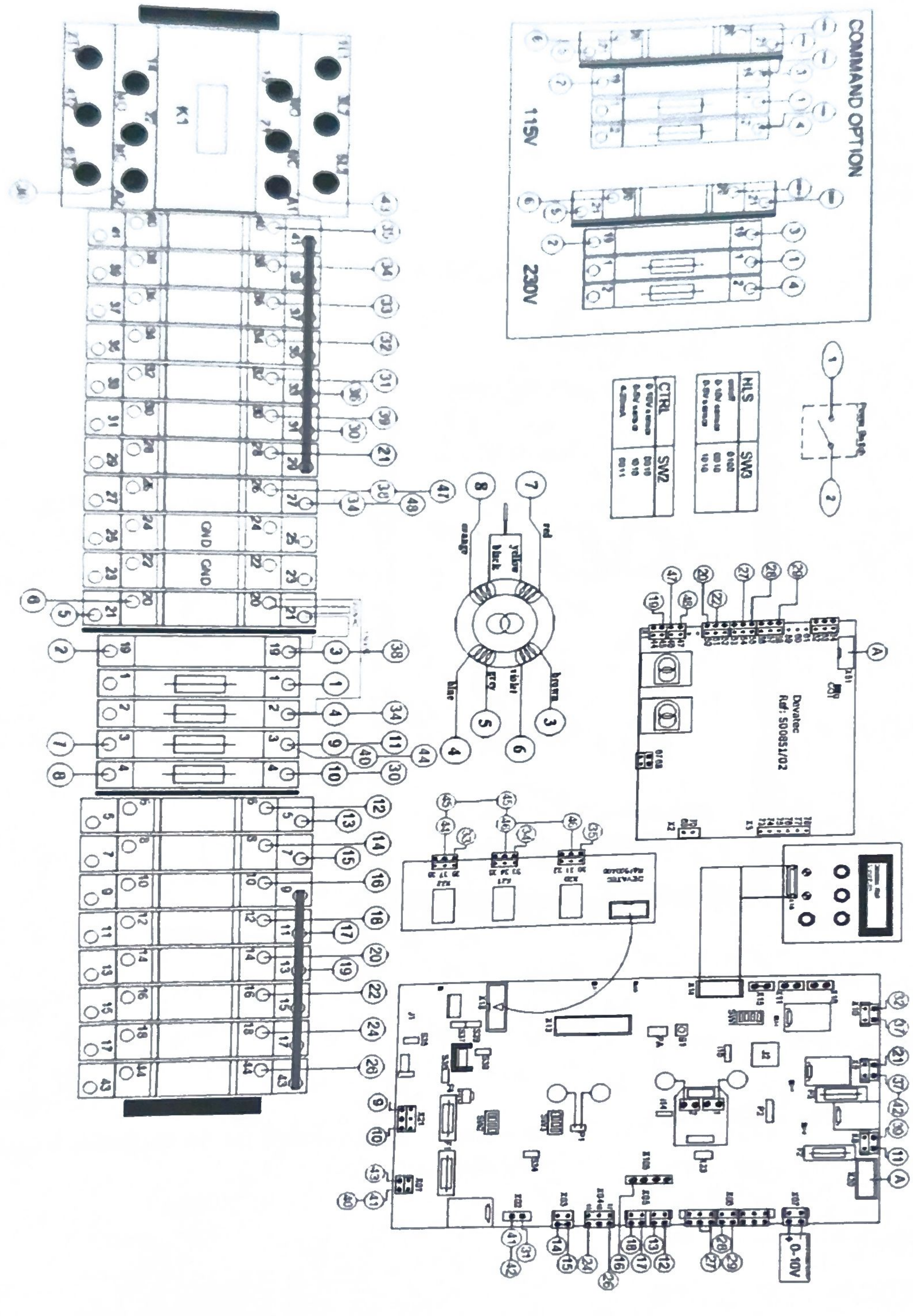


Option: Two temperature sensors can be installed (terminals 17-18 and 42-43) to read the air temperature at the inlet and outlet.

Option: Water supply renewal

	Option: This valve drains the water contained in the supply pipes, preventing stagnation and temperature rise during shutdown phases. The valve is installed close to the filling valve
	Hydraulic connection
	

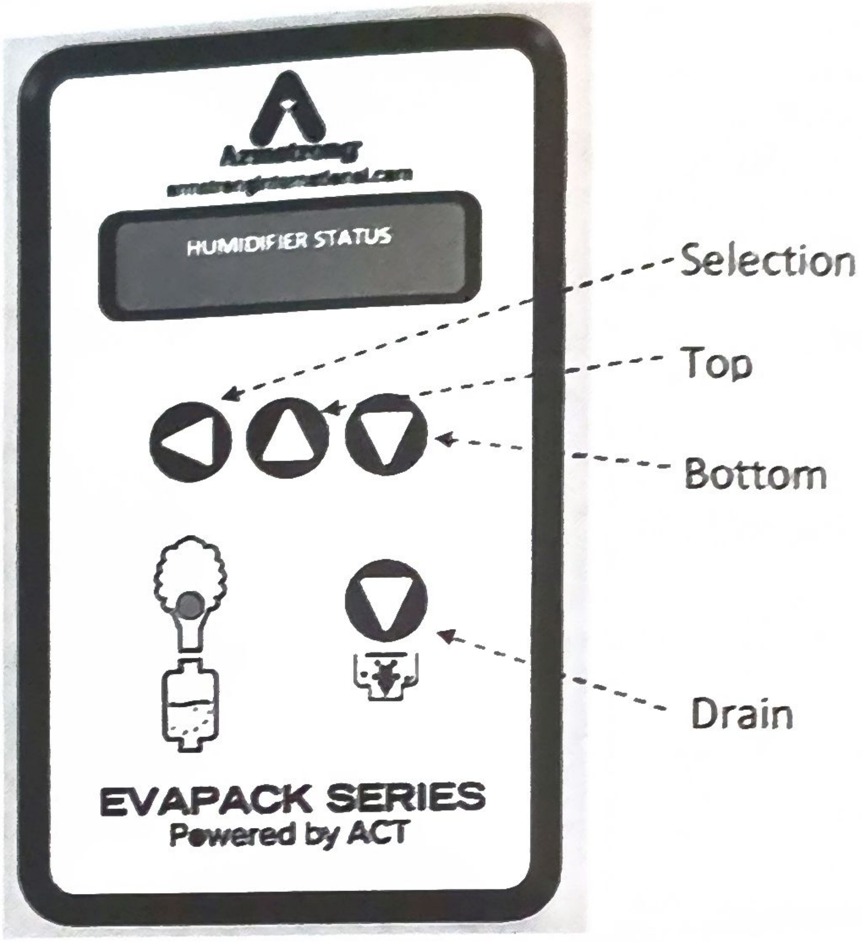
Wiring diagram



Commissioning

Before starting up the humidifier, check that the installation complies with the installation instructions.

- Switch on humidifier (power and control)
- Press ON/OFF switch
- The display shows "Filling in progress".
- Once the device has received the start-up command from the controller, the LED lights up.
- Press the selection button to navigate between the three main menus.
- To enter the menu, simply press the up or down button.



POWERED BY ACT

INITIALIZATION
EVP V XX.XX.XXX

HUMIDIFIER
STATUS

These menus appear quickly when the humidifier is switched on or in standby mode

Bleed-Off adjustment menu

This menu allows you to pre-adjust the duration and frequency of emptying according to water quality and basin size.
It only appears the first time the humidifier is switched on, and each characteristic must be entered before the humidifier can be operated correctly.

DATA FOR QUICK BLEED-OFF-SET-UP	
HARDNESS LEVEL: XXX	•Very Soft Water •Soft Water •Hard Water •Very Hard Water
BASIN WIDTH XXXmm / XXinch	**Basin Width: 500mm/20in - 600mm/24in ...** **... 2900mm/114in - 3000mm/118in**

→

Information menu (Read Only)

HUMIDIFIER STATUS	→ By pressing enter from this menu → HUMIDIFIER SETUP
INITIALIZATION EvP V XX.XX.XX	Code version
FILLING BASIN IN PROGRESS	Indicates if the basin is being filled.
REMOTE STOP	Device stopped by remote control (Modbus or Bacnet)
DUCT SAFETY SWITCH IS OPEN	Safety circuit open (terminals 7-8)
DEMAND XX %	Demand display (%)
VALUE OF CRTL XX V	With external control (0-10V, 4-20mA), displays signal value
ROOM Rh: XX% SET POINT: XX%	With a humidity transmitter, displays the relative humidity value measured in the room and the setpoint.
TOTAL RUN TIME xxxx hrs	Total operating time (hours)
TIME REMAINING TO EOL xxxx hrs	Hourly countdown to next maintenance service
SERVICE METER XXX hrs	Operating time before service
TIME REMAINING TO EOS xxxx hrs	This counter is activated when the unit is switched off. When the counter reaches 0 hours, the water in the basin is drained. EvaPack restarts when a new request is made.

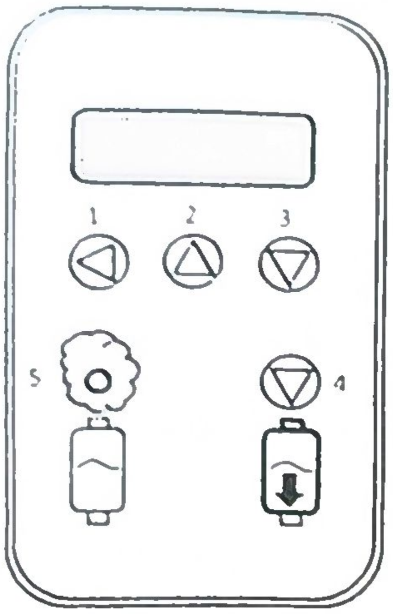
Setup menu

HUMIDIFIER SETUP → By pressing enter from this menu → **HUMIDIFIER CONFIGURATION**

CRTL SIGNAL TYPE XXX	Enter control type (ON/OFF CTRL, DIGITAL CTRL, 0-10V CTRL, 0-5V CTRL, 4-20mA CTRL, DIGITAL SENSOR, 0-10V SENSOR, 0-5V SENSOR, 4-20mA SENSOR)
SET POINT XX % RH	Appears if a sensor is used Enter the setpoint (1 à 99 % Rh)
PID SETTING P:xx I:xx D:xx	Adjust PID
END OF SEASON DRAIN? XXX	Complete basin emptying after a counter (YES/NO)
EOS TIMER SETTING XXX hrs	Indicates counter value (1 à 72 h)
WATER RENEWAL CYCLE XXX hrs	Complete renewal of the water in the basin after a period of operation (1 to 48 h)
LEVELS / FILL L:x H:x / x	Indicates the water level status (L: Low and H: High) (0: Level not reached, 1: Level reached) and inlet valve status (0: no filling, 1: filling in progress)
NWC HLS CRTL D% X X XXX XX	Indicates BMS digital control status (0 :OFF ;1 :ON), SLH servo status (0: open 1: closed), control signal CRTL and demand D%.
T1(°C) T2(°C) -X- -X-	Indicates temperatures measured by probes "X" indicates no probe (option)

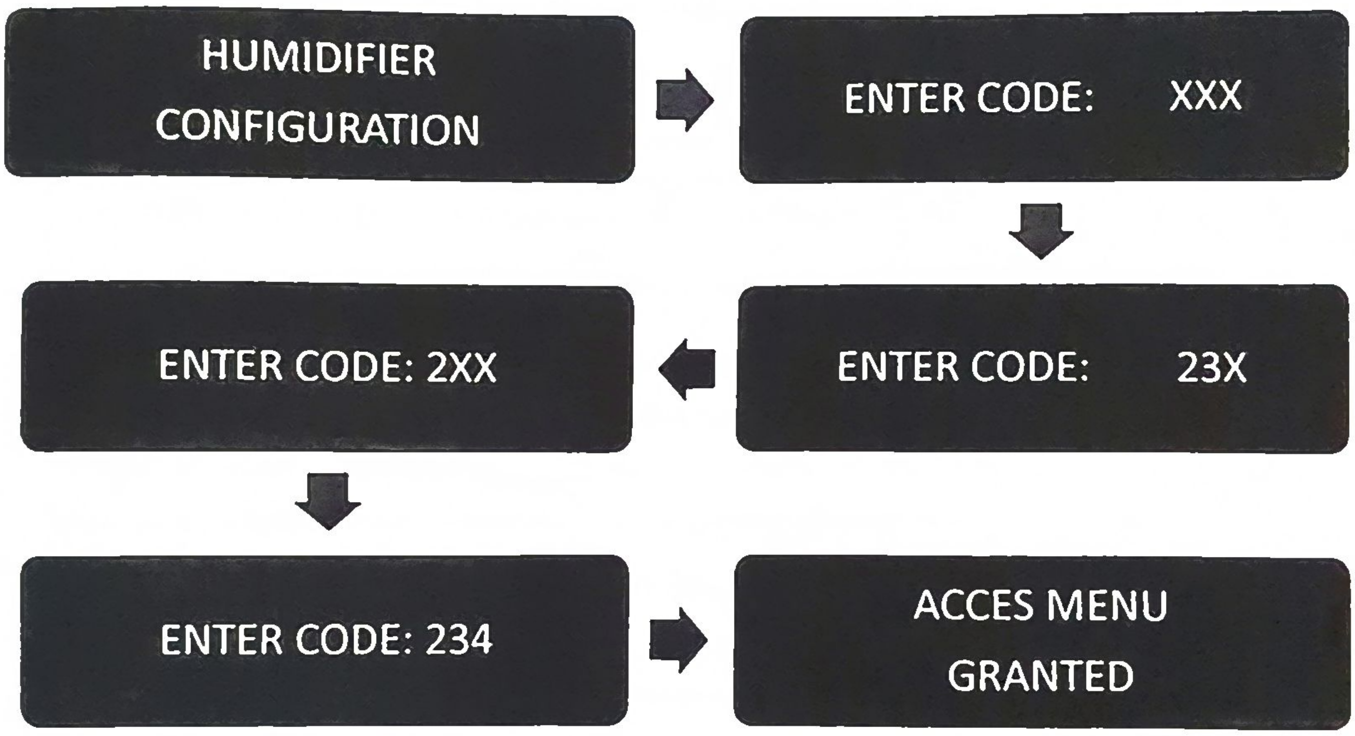
D% Pp V: 1 - 2 - 3 - 4	Indicates demand, pump status and stage control (0: OFF, 1: ON, X: Not used)
TIME TO CYCLE DRAIN min	Timer indicating operating time before draining (min) or draining time (sec)
DRAINING IN PROGRESS XX sec	Timer indicating the draining time [sec]
COM. DATA TX	Data Transmission (TX COM 1)
COM. DATA RX	Data Reception (RX COM 1)
NETWORK UNIT ID xx	Address ID setting (Modbus, 1 to 32, Bacnet 1 to 254)
COM SPEED	Speed protocol setting: Modbus RTU
COM PROTOCOL	Protocol settings: Modbus RTU

Menu change



How to enter your access code :

- Press key 1, the first cross flashes
- Press the 2 or 3 keys to change the number.
- Once you've reached the desired digit of your code, press 1 to confirm; the second cross will flash.
- Proceed in the same way for the following digits, and don't forget to validate your code by pressing 1.



HUMIDIFIER CONFIGURATION → By pressing enter from this menu → **HUMIDIFIER SETUP**

SELECT. LANGUAGE XXXX	Langue de navigation: FRANCAIS , ANGLAIS , etc...
EOL SETTING XXX h	Operating time before maintenance 100 to 20,000 hours
WATER TYPE USED XXXXX	Tap water, softened water, partial demineralized water, demineralized water (indicative) .
DRAIN LOGIC XXX	Allows you to manage the type of command for the drain valve ENERG.= DRAINING (with SV NC) ENERG.= NO DRAIN (with SV NO)
UNIT TYPE EVAPACK	Device type: EVAPACK
STAGED CONTROL XX	Number of stages in the control system (pump + valves: 1 to 5 stages) E.g.: 1 stage = pump; 2 stages = pump + 1 valve
CYCLE DRAIN DURATION sec	Drain time: 0 to 240 sec
CYCLE DRAIN FREQUENCY XX min	Drain frequency: 1 to 60 minutes
CALIBRATION FOR CTRL SIGNAL: %	Signal calibration (-10% to +10%)
CALIBRATION FOR T1 SENSOR %	Calibration of temperature signal T1

CALIBRATION FOR T2 SENSOR %	Calibration of temperature signal T2
CURRENT TIME XXXX/XX/XX XX/XX	Setting the date and time (to be reset after a power failure) Year/Month/Day Hour/Minute
TEST DATE XX/XXXX	Date du test usineFactory test date Month/year (read only)

Services and Maintenance

EOS DRAIN IN PROGRESS	The unit is stopped and the end-of-season counter has reached 0. Complete drainage of the unit is activated.
EOS REACHED DRAIN COMPLETED	When the basin is empty, the drain stops. The unit will refill when there is a new demand.
RENEWAL WATER IN PROGRESS XX sec	The basin is being filled with water
MANUAL DRAIN XXX sec	Draining: Press button (4), the message "MANUAL DRAIN" appears, the unit stops and draining is activated.
MANUAL DRAIN XXX sec	Once the basin has been emptied, the message "MANUAL EMPTYING COMPLETE" appears. Maintenance can now be carried out.

To reset the time counter to 0, press button (3) for 5s after the two messages.

EOL DRAIN IN PROGRESS	The unit is stopped and the end-of-life counter has reached 0. Complete emptying of the appliance is activated.
UNIT READY FOR MAINTENANCE	When the basin is empty, this message appears. Maintenance must then be carried out.

To reset this counter to 0, press button (3) for 5 sec after the two messages appear.

BLEED-OFF IN PROGRESS	**Option:** When the conductivity controller is present. If the conductivity of the water in the tank is too high, the drain valve opens until the conductivity returns to an acceptable level.

Troubleshooting

PUMP FAULT

__Meaning__

Alarm E0. Pump overheating detected. This indicates that the pump is running without water.

__In this case__ :

When this message appears, the tank is drained, the general fault contact is activated and the unit is then shut down.

E1: INITIAL FILL TIMEOUT

__Meaning__:

Alarm E1: During 1st filling, filling time too long (>40min)

E2: LOW LEVEL TIMEOUT

__Meaning__:

Alarm E2: Filling time to low level too long (>12min)

E3: LOW TO HIGH LEVEL TIMEOUT

__Meaning__:

Alarm E3: Filling time between low and high level too long (>12min)

E4: HIGH LEVEL TIMEOUT

__Meaning__:

Alarm E4: High level detection time too long.

__In this case__ :

Check water filling circuit, drain circuit and level detection, level detector, pump and stage valves

E5: INCORRECT LEVEL DETECTION

__Meaning__:

Alarm E5: High level reached (1) while low level not reached (0)

__In this case__ :

Check water level sensor